

Security Governance - Notes

CHAPTER 1: Introduction

The Enterprise as a Complex System

Every organization exists to fulfil a **mission**, that is, the fundamental business objective that justifies its existence and defines what it delivers to the world. An energy provider exists to produce and sell energy to its customers; a manufacturer exists to produce and distribute goods; a bank exists to safeguard and move money. This mission is the anchor from which every other organizational choice descends, because once the objective is fixed, the enterprise must structure itself in a way that allows that objective to be reached reliably and repeatably.

From the outside, an enterprise can be understood as a kind of black box: it consumes inputs and produces goods or services that are offered to customers. Inside that black box, however, sits a highly structured machine composed of three interlocking ingredients. The first ingredient is **people** — the employees whose work actually moves the organization toward its objective. The second is **processes** — the structured sequences of activity through which work is carried out, formalized under the discipline of Business Process Management (BPM), in which people and technology transform inputs into outputs. The third is the **digital infrastructure** — the technological substrate of servers, networks, applications and data on which the people rely to perform their processes. The way these three are arranged depends heavily on scale: a small company tends to have a flat structure with few intermediaries, whereas a large enterprise develops a deep hierarchy with many layers of coordination.

This composition is precisely what makes the enterprise a **complex system**, and it is also the source of its first and most persistent vulnerability. Because the digital infrastructure is operated by human beings, the point of contact between people and technology becomes the weakest link in the whole arrangement. Managing the security of even a small company is therefore never a trivial task, and the difficulty grows sharply with organizational size and complexity.

The Concept of Governance

Among the many processes an enterprise runs, this treatment concentrates on **governance**. Governance, in its broadest sense, is the totality of the processes of

governing — whether these are undertaken by a government, a market, or a network, and whether they act over a family, a tribe, a formal or informal organization, a territory, or across territories. Governance operates through the laws, norms, power, and language of an organized society. Expressed in operational terms, governance is the set of processes of interaction and decision-making among the actors involved in a collective problem, and these processes lead to the creation, reinforcement, or reproduction of social norms and institutions.

Two words distill the essence of this definition and should be held firmly in mind: **processes** and **norms**. To govern is to look at the business objective and to put in place everything necessary to reach it, which means identifying the processes required and then operating those processes so that the objective is achieved *in compliance with the surrounding environment*. Doing governance therefore always carries a double burden. It is not enough to design processes from a purely functional point of view, asking only whether they work; one must simultaneously keep the **compliance** dimension in view, ensuring that the processes respect the applicable norms. Those norms may be external laws, internal company rules, contractual obligations, or regulatory prescriptions. Governance without compliance is incomplete governance.

Enterprise, Corporate, and Business Governance

The most general form of this discipline is **enterprise governance**, which concerns how to govern the company as a whole. Formally, enterprise governance is the structure and the set of relationships that control, direct, or regulate the performance of an enterprise together with its projects, portfolios, infrastructure, and processes. In other words, it is the exercise of identifying the processes necessary to control the company itself. Enterprise governance is conventionally split into two complementary halves.

The first half is **corporate governance**, understood as the system of rules, practices, and processes by which a firm is directed and controlled. Corporate governance is essentially concerned with **compliance processes**: it is a general, mission-independent approach that guarantees the firm is being run in a controlled and accountable way, and it applies to any company regardless of what that company actually produces.

The second half is **business governance**, understood as the set of policies and business processes that determine the way the organization's business is actually run. Business governance is concerned with **performance processes**, and unlike

corporate governance it is highly specific to the mission. Where corporate governance asks whether the firm is being directed and controlled properly in a generic sense, business governance asks whether the firm is performing well against its particular objective. The two are therefore distinguished cleanly: corporate governance is general and compliance-oriented, business governance is specific and performance-oriented.

The Hierarchical Structure and the Direct/Control Loop

An enterprise is typically organized as a **pyramid**. At the apex sit a small number of people who make broad, high-consequence decisions; those decisions then propagate downward through the structure, becoming progressively more concrete at each descent. This vertical arrangement gives rise to three well-known management levels, and the mechanics of corporate governance can be read directly off them.

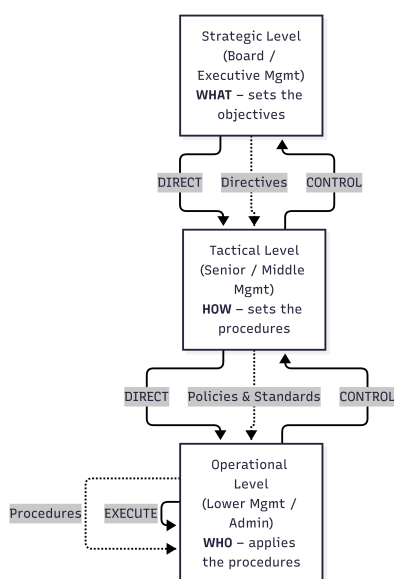
At the **strategic level** — populated by the Board of Directors and executive management — decisions about cybersecurity are taken at their most abstract. Directors formalize these decisions into **Directives**, documents that state the strategic objectives of the company. A directive at this altitude might declare that the infrastructure must be protected from external attackers, but it deliberately refrains from specifying *how* this is to be accomplished. The strategic level owns the **WHAT**: it sets the objectives.

At the **tactical level** — populated by senior and middle management — the mandate received from above is translated into more concrete objectives. Here the discussion turns to *how things should be done*, and the abstract goal begins to acquire shape. It is at this layer that one starts to identify the procedures needed to protect the assets, defining the **policies** and the **company standards** that will govern behaviour. These standards are not necessarily imposed from outside by bodies such as ISO; they are frequently the internal standards the company chooses to adopt for itself. The tactical level remains a *meta* layer, in the sense that it defines the way things must be done without yet touching the machines. The tactical level owns the **HOW**: it sets the procedures.

At the **operational level** — populated by lower management and administration — the middle-level objectives are finally converted into practical reality. Here the abstract instruction to deploy, say, an identity management system becomes a concrete decision about who does exactly what, on which machine, with which

configuration. This is the layer in which people put their hands on the systems. The operational level owns the **WHO**: it applies the procedures, and it is where things are actually done.

These three levels are bound together by a continuous loop composed of three motions. **Direct** is the downward flow: it translates big, abstract objectives into concrete implementation procedures as they descend from strategic to tactical to operational. **Execute** is the practical act of doing the work at the operational level. **Control** is the upward flow: it observes whether what is being done actually works as expected, and it feeds the resulting information back up to management. When the feedback arrives at the top, management obtains a snapshot of the real situation — an understanding of how the company is genuinely behaving — and on the basis of that snapshot it can decide to hold course or to change direction, re-drafting the objectives as needed. The following diagram captures this continuous circulation.



The single most important idea to extract from this picture is that **talking about governance means talking about the implementation of a continuous loop**. Governance is not a one-off act of decision but a perpetual cycle of directing, executing, controlling, and re-directing.

Vulnerabilities as the Enemies of Governance

Once this loop is running inside a real company, it must contend with **vulnerabilities**, which are the natural enemies of the governance process. Vulnerabilities arise for two broad reasons. The first is that the technical infrastructure itself changes over time and inevitably contains bugs, errors, and even backdoors. The second, and more insidious, is the interaction between the

processes and the infrastructure — and, above all, the interaction between *people* and the infrastructure. It bears repeating as a central lesson that the most dangerous moment in the whole governance enterprise is the moment at which people come into contact with the systems.

Risk as the Governing Metric

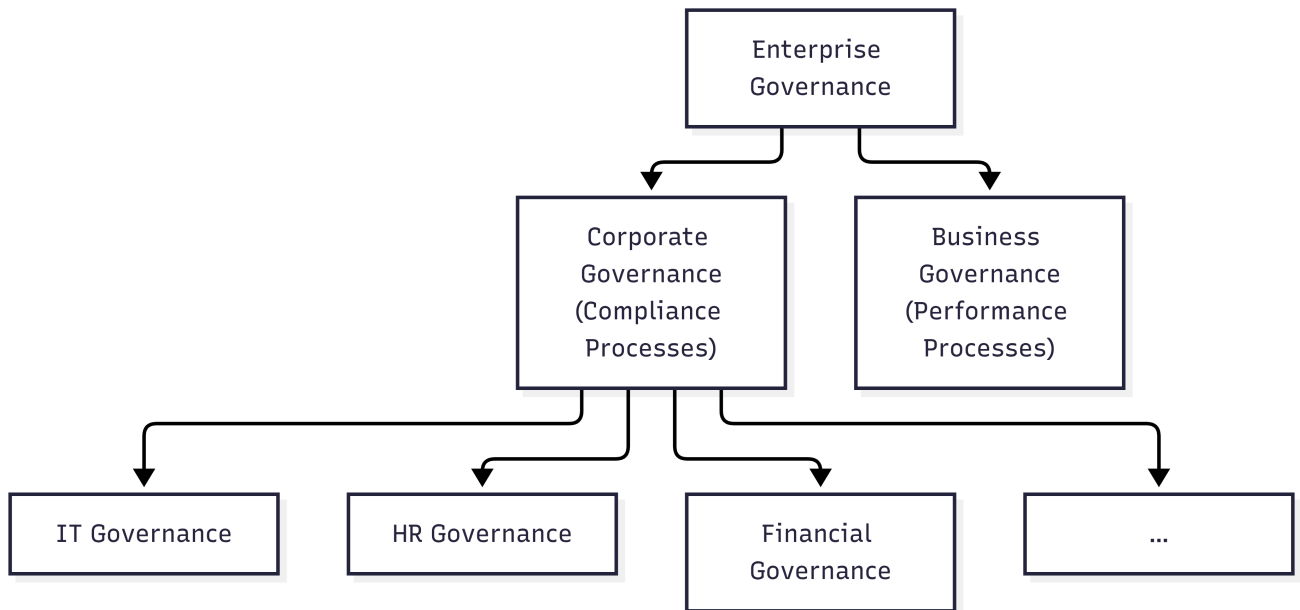
If the direct/control loop is to be steered intelligently, it must be steered against a measurable quantity. That quantity is **risk**, which may take financial, production-related, cybersecurity, or other forms. The IT Governance Institute captured this centrality when it redefined enterprise governance as the set of responsibilities and practices exercised by the board and executive management with the aim of providing strategic direction, ensuring objectives are achieved, ascertaining that risks are managed appropriately, and verifying that the enterprise's resources are used responsibly.

Once risk is accepted as the governing metric, the entire governance problem collapses into a clean **optimization process**. The direct half of the loop is charged with a single objective: reduce risk, by putting in place actions that lower the risk level. The control half of the loop is charged with the complementary objective: measure the reduction, by taking measurements, recomputing the risk, and aggregating it into a coherent picture. Governance thus becomes the continual minimization of risk under observation.

Decomposing Governance: IT Governance and Assets

Supporting business and corporate governance in practice requires more specialized sub-governances, each dedicated to a particular domain of the enterprise. Enterprise governance decomposes into corporate governance and business governance at the top, and beneath these sit the domain-specific governance sub-systems: **IT governance**, **HR governance**, **financial**

governance, and others. The following diagram shows this decomposition.



IT governance is defined as the set of processes that ensure the effective and efficient use of IT in enabling an organization to achieve its goals. It begins from the infrastructures and examines every kind of risk that arises from the digital environment, and especially the risks that arise from the *offering of services*. IT-based systems create serious risks precisely because they manage and store the majority — if not the entirety — of a company's electronic assets: the data and information stored inside the organization, the data and information transmitted across its networks, and all the systems and applications required to store, transmit, and process that data. These electronic assets are exposed to a wide range of threats aimed at compromising their three cardinal properties, **Confidentiality, Integrity, and Availability**, collectively abbreviated as **CIA**.

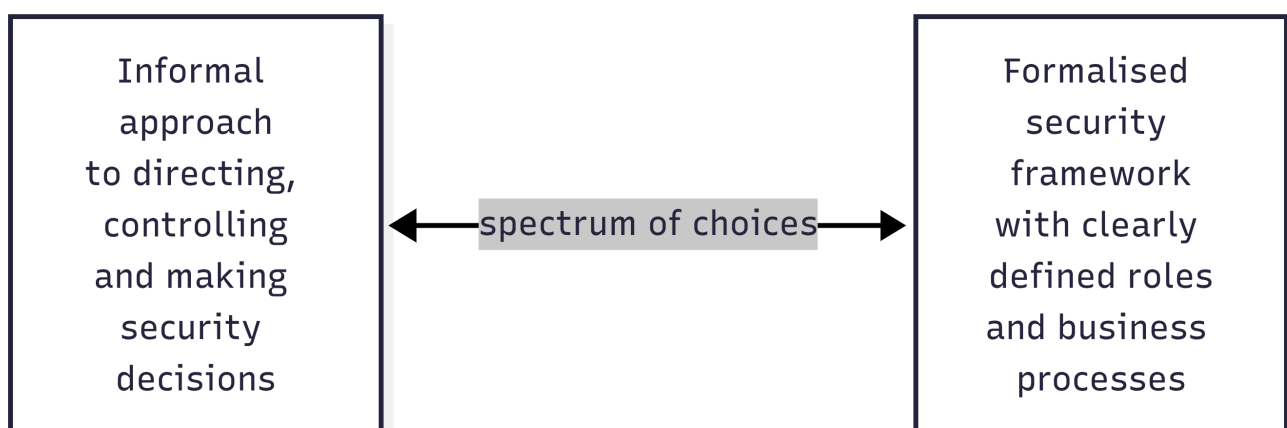
The decisive insight of IT governance is that the infrastructural view must be fused with the business view, because the two views assign radically different importance to physically identical machines. Consider a network of interconnected servers. On one server sits an archive of data; on another, identical machine sits the web server that hosts the company's online shopping site. Viewed through the infrastructure lens alone, the two servers look equivalent — both are connected, both are running, both are simply nodes in the topology. The moment the business lens is applied, the equivalence collapses. If an attacker brings down the shopping site, the company can no longer serve its customers and immediately begins to lose money, which is a serious risk. If an attacker damages the archive, the data can very often be restored from other sources, which is a comparatively minor risk. The two machines are the same; the risk they carry is not. This is why IT governance insists on considering *all* aspects of the digital world — not merely the

machines, but everything operating around them — and on first identifying which assets genuinely matter. Running a live service is an asset in the critical sense; the archive, in this example, is not an asset of the same order.

The threat landscape, moreover, is not confined to the outside. One must protect against external attackers arriving from the Internet, but one must equally guard against attackers already *inside* the organization. Employees who have become unhappy are a particularly grave danger: they know the assets intimately, they understand how the infrastructure is managed, and this superior knowledge makes them far more dangerous than an outsider who must first discover what an insider already knows.

Choosing an Approach: The Spectrum of Formalization

Having established that governance is a risk-driven control loop operating over identified assets, the practical question becomes how to implement that loop in a concrete company. There is no single universal answer — there is **no "one size fits all" approach** to security governance. Instead there is a spectrum. At one extreme lies a fully **formalized security framework**, with clearly defined roles and rigorously defined business processes. At the opposite extreme lies an **informal approach** to directing, controlling, and making security decisions. Every real company must find its proper place somewhere along this continuum.



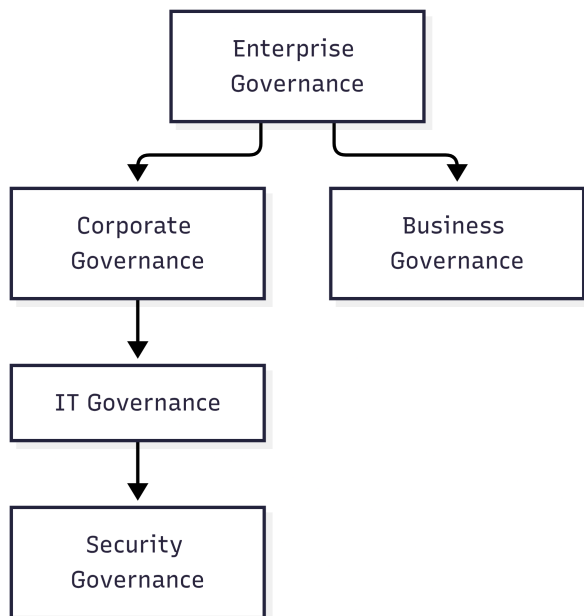
Four questions guide a company toward the right level of formalization. The first concerns **size and complexity**: how large and how complex is the organization? Small companies must lean on flexibility because they typically lack the resources for heavy formalism, while large companies will most probably require a formalized security framework simply to keep coordination tractable. The second concerns **resources**: what budget is actually available for security governance? A formalized approach is expensive — it demands certifications and sustained investment — so

a company without money cannot realistically adopt it. The third concerns **mission and the importance of security to it**: what does the organization do, and how important is security to those aims? For a bank, security is a top priority and a matter of existential dependence; for a manufacturer, security still matters but the business does not hang on the same human-facing service dependencies. The fourth concerns **external considerations**: are there contractual, legal, regulatory, or sector-specific requirements pushing the company toward a particular level of formalism? A small company might in itself prefer an informal posture, yet find itself obliged to obtain certification because it works for a public organization that demands it.

Answering these questions well requires analysing the **context**, and in practical terms the correct approach comes down to identifying three things. One must identify the **what**, meaning the security decisions to be made. One must identify the **who**, meaning the deciders who will make them. And one must identify the **data**, meaning the information required to make sensible and informed choices. These three together — decisions, deciders, and the information feeding them — constitute what is meant by a **security framework**.

Positioning and Purpose of Security Governance

Security governance can now be located precisely within the larger structure. It is the means by which an organization controls and directs its approach to security, and when it is done well it coordinates all of the organization's security activities and enables the flow of security information and security decisions throughout the organization. Structurally, security governance is a **vertical** concern that cuts across the other governances: it sits within IT governance, which sits within corporate governance, which — alongside business governance — sits within enterprise governance. The nesting looks as follows.



The reasons to care about security governance follow directly from this position. It provides a structured approach to managing and mitigating cybersecurity risks within the organization. It helps protect sensitive data, maintain regulatory compliance, and strengthen the trust and confidence of stakeholders. And by implementing effective security governance, an organization reduces both the likelihood and the impact of security incidents, thereby safeguarding its reputation and its business operations.

This defines the arc of the entire subject. The study of security governance will proceed through models for information security governance and the vertical nature of security; through cybersecurity best practices, guidelines, and frameworks and how to use them; through the assessment and management of risk, which entails understanding threats, vulnerabilities, and attacks, evaluating their impact, and finding proper countermeasures; through the modelling and understanding of threats; through the management of incidents; and finally through legislation, compliance, and insurance, closing on open research issues in security governance.

A Model for Information Security Governance

Recap: What Information Security Governance Is

Before assembling the full model, it is worth restating the object of study with precision. **Information Security Governance (ISG)** is the process of establishing and maintaining a framework, together with the supporting management structure and processes, that provides assurance of several things at

once: that information security strategies are aligned with and support the business objectives; that they are consistent with applicable laws and regulations through adherence to policies and internal controls; and that responsibility is properly assigned throughout the organization — all of this undertaken in an effort to manage risk. Said more plainly, ISG is the global effort a company makes to ensure the well-being of its electronic resources.

The Principle of Cost-Effectiveness

The first foundational principle on which any ISG framework must be built is that it has to be **cost-effective**. Security is expensive, and there must therefore be a deliberate balance between the cost of protecting electronic resources and the risk to which those resources are exposed. This balance has two failure modes, each equally to be avoided. **Overprotection** wastes money on defences whose cost exceeds the harm they prevent, generating unnecessary expense.

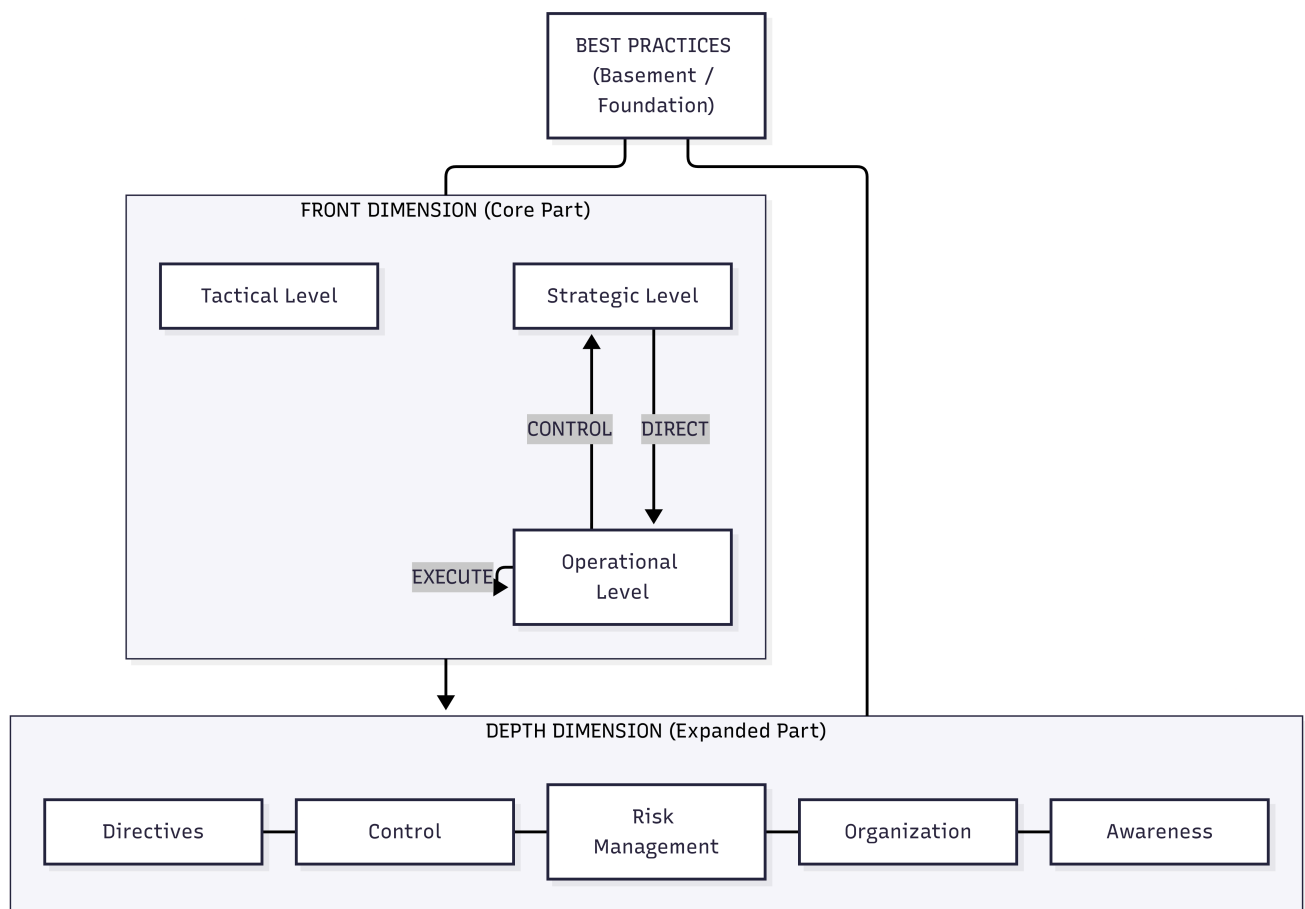
Underprotection leaves genuine exposures unaddressed, allowing risks to materialize and strike the company. Good governance lives in the narrow band between these two errors.

The practical force of this principle becomes visible the moment one imagines the two possible worlds. Suppose a risk assessment has been carried out and has produced a list of risks, each tagged with an associated security level. In an idealized world with no budget limit, one could simply march down the list and remediate every risk one by one until none remained. Real organizations do not live in that world: they operate under a **fixed budget** and must decide where that budget goes, starting from the most critical risks and working downward. A characteristic situation then arises in which most of the budget has been consumed fixing the critical problems, and a residual amount remains. At this juncture the decision must be made carefully, weighing the cost of fixing a given remaining risk against the cost that would be incurred if that risk actually materialized — neither overspending on marginal exposures nor abandoning issues that quietly matter. Even the act of *identifying* a risk depends on acquiring and analysing data with the right instruments; if the proper instruments are not used, important points may simply be missed, and the balance will be struck on incomplete information. For all these reasons, a good **IT risk management strategy** is mandatory as the precondition for a good ISG strategy, and good governance as a whole demands a strategy that weighs multiple aspects rather than fixating on the infrastructure alone.

The Von Solms Model: Overview

The model used to organize all of these elements was introduced by **R. Von Solms in 2006**. It can be pictured as a solid three-dimensional wedge whose purpose is to gather every element one must keep in mind and to arrange those elements in an interconnected structure. The wedge has three distinguishable parts, and understanding the model means understanding what each part contributes.

The **Front Dimension**, also called the **Core Part**, is the triangular face of the wedge. It is the implementation of the control loop already studied, spanning the three management layers of the pyramid. The **Depth Dimension**, also called the **Expanded Part**, is the set of five layers extending backward from that front face: **Directives, Control, Risk Management, Organization, and Awareness**. Finally, the **basement** on which the whole wedge rests is the **Best Practices** layer, the foundation from which everything else is constructed. Best practices are the documents that collect the common, accumulated knowledge of the field in terms of cybersecurity and management. The overall structure is shown below.



The Front Dimension (Core Part)

The Front Dimension represents the execution of processes and actions and the influence of the Direct and Control loop upon those processes. It rests on two core principles.

Core Principle 1 states that the model covers the three well-known levels of management — strategic, tactical, and operational. This principle answers the question of *why all three levels must be covered by the governance process*: every level possesses its own distinct view of the company, and no single level can see the whole. At the top, a handful of people take general decisions and, when the direct process begins, they possess only ideas about what to do; the directive documents they eventually write are therefore quite general. At the bottom, people are in direct contact with the system, laying their hands on machines and producing procedures, and their scope is extremely detailed and correspondingly restricted. Because each person's view is limited in this way, the intermediate tactical layer is indispensable: its whole function is to translate the general into the detailed on the way down, and, on the way up, to translate the detailed back into the general. This is why the processes must be able to flow *vertically* in both directions. On the control path, the operational level begins by collecting information of great precision — numbers of alerts, flows across different machines, and similar indicators — and this precise information is progressively correlated with the procedures and abstracted as it climbs, so that the movement upward is a movement from a very detailed view toward an increasingly summarized one.

Core Principle 2 states that across these three levels there are three very distinct *actions*, and that control at the bottom is emphatically not the same activity as control at the top. The rationale is easy to appreciate from the Board's perspective: the Board is not interested in learning that 321 virus attacks occurred last week, that 27 software patches were applied, and that 7 anti-virus updates took place — this is far too much detail to plan with. What the Board wants to know is whether the company's access to the Internet causes any *unmitigated* risks to the company. If there are none, the Board is content; if there are some, the Board wants to be told precisely which unmitigated risks these are. The action of control must therefore be qualitatively different at each altitude.

The Direct Part, Layer by Layer

The downward Direct process can be read as a matrix in which each management layer has a distinct meaning, a specific input, and a specific output, and in which the output of one layer becomes the input of the next.

At the **strategic level**, to direct means to **identify the assets, their relevance, and their required level of protection** — appropriate here because the strategic level alone possesses the overview of the whole company and can therefore say what must be protected and to what extent. Its **input** is the

context, split into internal and external factors. The internal factors concern the company's own world: its strategic vision, the role of IT, its competitiveness, its mission, its constraints, and its production processes. The external factors concern the surrounding world: legal and regulatory prescriptions and other external risks. Its **output** is a set of **Directives** indicating, at a high level, what the Board expects to be done as far as the protection of the company's information assets is concerned.

At the **tactical level**, the directives received from above are *expanded* into sets of relevant information security **policies, company standards, and procedures**. Here the macro-objective is analysed against the situation and translated into middle-level objectives. The **input** is the set of directives produced above, and the **output** is the collection of policies, procedures, and standards with which the company must comply — the way digital resources are managed, the way accounts are handled, and so on.

At the **operational level**, the inputs are expanded further into sets of **administrative guidelines and administrative procedures**, and the technical measures are physically implemented and managed. The **output** is the set of **operating procedures** specifying exactly how things must be done, and this output forms the very basis of execution on the lowest level.

The Control Part, Layer by Layer

The upward Control process is governed by a single, decisive maxim: **you can only manage what you can measure**. To control properly one must measure; to measure one must know in advance which information and data to collect. This *measurability* characteristic cannot be an afterthought — it must sit at the centre of all directives, policies, standards, and procedures produced during the Direct part of the model, so that the objects being directed are constructed to be measurable in the first place. The Control matrix mirrors the Direct matrix but runs in the opposite direction, from operational up to strategic.

At the **operational level**, to control means that **measurement data is extracted from a wide range of entities**, either automatically or manually. Concretely, this means deploying monitors — **agents** — into the environment to collect and extract data. These agents may live on the machines themselves or in the network, and their power depends on their observation range. Here a fundamental tension must be respected: **the need for monitoring must not compete with the needs of the business**. Observing everything by placing

agents everywhere is expensive and degrades the system's speed; when one interferes with the system to monitor it, one cannot do too much, or the system will suffer. The guiding rule is that the need for protection must exist and survive alongside the need of the business, without smothering it. The **input** at this level is the raw **measurements**, and the **output** is a set of **specialized reports** created on this level from the extracted operational data, which is then passed upward to the tactical level.

At the **tactical level**, control means **measurement and monitoring against the requirements of the relevant policies, procedures, and standards**. The tactical layer receives all the operational reports, aggregates them together, and attempts to identify existing threats, evaluating whether the policies in force are sufficient. Its **output** is a set of **tactical management reports** indicating the levels of compliance and conformance, which are then encapsulated and passed to the strategic level.

At the **strategic level**, control culminates in **Situational Awareness**. The **output** at this apex consists of **reports reflecting compliance and conformance to the relevant directives, including risk considerations** — the very snapshot the Board needs in order to decide whether to hold course or re-direct.

The Basement: Best Practices

Underneath both the Front and the Depth dimensions lies the layer of **best practices**, which sets the ground on which all operations are instantiated. Best practices are documents collecting the knowledge that people have acquired over the years in terms of security and management. The specific best practice one reaches for depends on the specific problem to be solved, and the required granularity depends on the management level being served.

The purpose of the whole ISG model, phrased memorably, is to help the organization in *"doing the right things right."* This slogan resolves into two questions that every **Information Security Manager (ISM)** must be able to answer. First: *how do I know what the right things are?* Second, supposing that is known: *how do I know I am doing them right?* The answer to both is: look to best practices. Formally, best practices — also called standards or guidelines — are a set of documents reporting the experiences and solutions arrived at by experts in information security management. They can be understood as the consensus of experts in the field, and they provide an internationally accepted framework usable

as a building block for ISG. Many of them exist, each focused on related but different aspects, including the **ISO 27000 family**, **COBIT**, the **NIST SP 800-* series**, the **CSC** (Critical Security Controls), and the **NIST CSF** (Cybersecurity Framework), among others.

ISO 27002

Among these best practices, **ISO 27002** deserves close attention because companies use it to set up their infrastructures from the policy point of view. It is an International Standard designed to be used by organizations that intend to do any of three things: to select controls within the process of implementing an Information Security Management System (ISMS) based on ISO/IEC 27001; to implement commonly accepted information security controls; or to develop their own information security management guidelines.

ISO 27002 is fundamentally a collection of **controls** — points that need to be verified. In its 2013 edition it contains **14 security control clauses**, which collectively hold a total of **35 main security categories** and **114 controls**. (These controls are general security controls and are not themselves objects of certification; they function as instructions rather than as a pass/fail certificate.) The structure is strictly hierarchical: each clause that defines security controls contains one or more main security categories, and each main category in turn contains two things — first, a **control objective** stating what needs to be achieved, and second, one or more **controls** that can be applied to achieve that objective.

The description of each individual control is itself standardized into a fixed structure of three parts. The first part is the **Control** proper, which defines the specific control statement needed to satisfy the control objective. The second part is the **Implementation Guidance**, a longer text that provides more detailed information to support the implementation of the control and the meeting of its objective; this guidance carries an explicit caveat, namely that it may not be entirely suitable or sufficient in every situation and may not fulfil an organization's specific control requirements. The third part is **Other Information**, which provides further considerations such as legal factors and references to other standards, and which is simply omitted when there is nothing further to say. In this way a control functions as a security checklist item that matches the requirements for certification, accompanied by a guideline explaining one possible way to implement something compliant with the check.

ISO 27001 and the Distinction Between the Two Standards

ISO 27001 is the companion International Standard that specifies the requirements for establishing, implementing, maintaining, and continually improving an ISMS within the context of the organization. It also includes requirements for the assessment and treatment of information security risks tailored to the organization's needs, and it is deliberately generic, intended to be applicable to all organizations regardless of type, size, or nature. Crucially, ISO 27001 is the standard that grants **certification**: it certifies that a system is compliant with its requirements, and it operates as a checklist of security checks that a company must satisfy to be formally certified.

The relationship between the two standards is best understood as a division of labour, and the difference between them is the difference between a requirement and a guideline. **ISO 27001** is very specific and strict; it spells out in detail what a company must comply with and must have in place in order to be formally certified. **ISO 27002**, by contrast, is a *guideline* document. It advises companies on what they should have in place as far as their information security management is concerned in order to follow best practice, guiding a company to structure its information security management according to the accumulated experience of other companies. It is deliberately rather high-level and does not drill down into very detailed specifics. The compact way to remember the pairing is that **27001 certifies and is strict, while 27002 guides and is advisory**.

The Directive and the Information Security Policy Architecture (ISPA)

Why a Documented Direct Process Is Needed

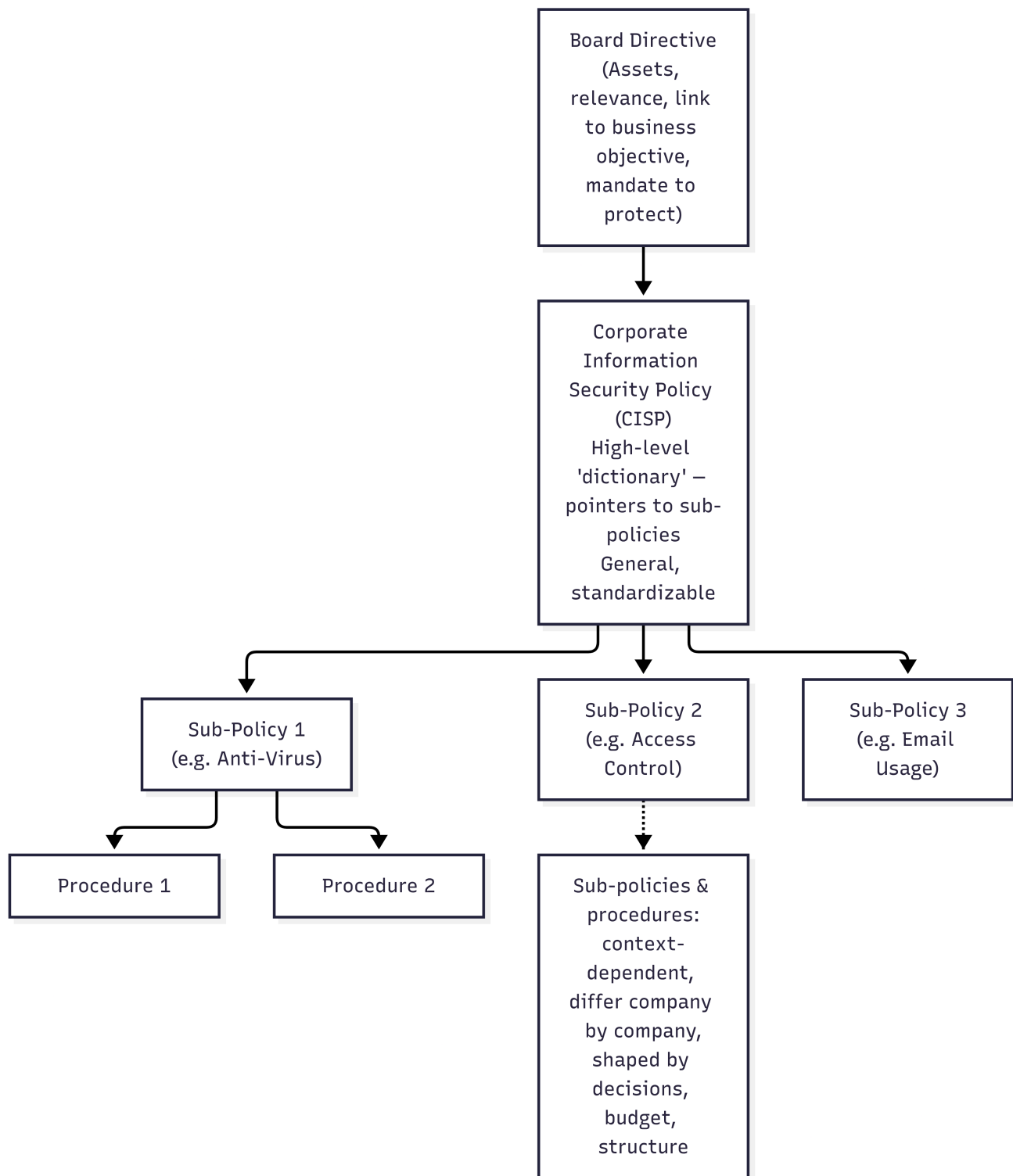
Every best practice contains, in some form, the notion of a **Directive**, and the task it imposes is concrete: to set up the documents and policies that dictate how security is to be governed inside a company. Chapter 5 of ISO 27002, for instance, requires that decisions be taken and acted upon in a *structured* way, which in effect demands the design of a process that supports all three management layers. To meet that demand, an organization must develop a **framework** — a methodological instrument, a defined set of things that must be done — capable of supporting the Direct process. This framework is not a single document but an

interconnected architecture of documents, each linked to the others, which together implement the guidelines of the downward direct flow.

The reason everything must be written down is that documentation is what confers **trust and value**. By formalizing rules and policies on paper, the organization gives weight and accountability to what it does; an undocumented intention carries no assurance, whereas a documented one can be communicated, enforced, audited, and relied upon. Because the Direct process flows from top to bottom, and because each layer operates at a different degree of abstraction, the architecture must provide a set of documents calibrated to the right level of information at each layer. A general document is produced at the top and is connected downward to progressively more specific documents produced in the layers beneath it.

The Document Tree

The architecture of the Direct process is best visualized as a **tree**. At the root sits the **Board Directive**. One level below sits the **Corporate Information Security Policy (CISP)**, which receives the directives from the board and transforms them into guidelines for the implementation of security controls. From the CISP, a set of **sub-policies** branches out, one for each distinct domain of concern — access control, human resource management, device management, and so on. From those sub-policies are derived the **company standards**, which express the minimum level of specification that must be achieved, and finally, at the operational layer, the **procedures**, which are protocols: ordered lists of concrete actions. The flow and its contents are shown below.



A crucial distinction runs through this tree. The **Board Directive** and the **CISP** are relatively **general and standardizable**, because almost every company shares essentially the same fundamental need for protection, and so the upper two documents can look broadly similar across organizations. The **sub-policies** and the **procedures**, by contrast, are strictly **context-dependent**: they vary from company to company and are shaped by the particular decisions the organization wishes to enforce, by its structure, and by the budget it has available.

The Board Directive

At the top of the tree, the Board Directive carries the raw material of the entire Direct process. It must contain two things above all: the **set of IT assets identified by the Board**, together with their relevance and their connection to the business objective, and the **mandate from the Board to protect those assets**. The inclusion of relevance is essential, because protection must proceed *asset by asset* — not every asset carries the same importance, and the directive must make those differences explicit so that protection effort can be allocated accordingly.

The Corporate Information Security Policy (CISP)

One level lower sits the CISP, the **high-level document that provides the basis for all lower-level documents related to information security**. Its function is to take the objective expressed in the Board Directive and translate it into a middle-layer objective, and from that translation it extends a set of pointers aimed at the specific sub-policy documents below. In this sense the CISP operates as a kind of **dictionary**: it allows the reader to interpret and map the sub-policies against the original board directive, binding the detailed and the general together.

The methodology offers a set of guidelines for writing a proper CISP, specifying both what it should contain and how it should be formalized. There are ten such guidelines.

First, the CISP must indicate **board and executive management support and commitment**, and it must be clear that the CISP flows from a higher-level directive; the link back to the Board Directive and the enforcement behind it must be visible. Second, it must be **accepted and signed by the CEO or an equivalent officer** — the signature of a board member is what gives the document its authority. Third, it **must not be a long document, nor written in technical form**: a maximum length of about four to five pages is appropriate, and it should contain high-level statements concerning information security rather than implementation detail. Fourth, it **should not change very often and must be stable** with respect to technical developments and changes; because it is an intermediate working document bridging the direction and the operational levels, frequent revision would destabilize everything hanging beneath it. Fifth, and as a direct consequence of the stability requirement, the CISP **must not contain any references to specific technologies and must be technology-neutral**, since technology evolves far too frequently to be enshrined in a stable document.

Sixth, the CISP must indicate **who owns the policy** and what the responsibilities of other relevant people are; stating roles and responsibilities is what creates a structure that links each action to a specific person, producing a map of who is supposed to do what and who is supposed to monitor it. Seventh, it must clearly indicate the **scope** of the policy — that is, all the people who will be subject to it. Eighth, it must refer to possible **disciplinary actions for non-conformance**, both to the CISP itself and to its lower-level constituent policies; these disincentives are necessary precisely because of the human factor, since people tend to look for an escape strategy when confronted with something difficult. Ninth, it must be **distributed as widely as possible** throughout the company and covered in all relevant awareness courses, so that employees understand both the policies and the reasons they exist. Tenth, it must contain a **Compliance Clause**, the element that ties the document directly into the Control process.

A representative CISP is organized into clearly named but deliberately general sections — the importance of information to the company, what information security is, why the company needs it, the information security commitment of the board, the scope, the information security statements that make up the policy, the responsibilities, and the compliance clause. The document is short, and its section titles are generic. Where it does become concrete is in the **scope**, which must define exactly who is affected by the policy: typically all of the company's IT assets stored, processed, and distributed via its information processing systems; any person granted authorization to access those resources, including permanent, temporary, third-party, and contractual employees and users; and all business partners and clients that access the company's IT assets in any way.

Sub-Policies

Descending one further level, the **sub-policies** take the mandate to protect the assets — already translated into macro-objectives by the CISP — and define specific policies for the specific aspects that must be considered. These documents are **company-dependent**: each organization must define them according to its own structure and circumstances. Nevertheless, certain categories of sub-policy apply to essentially everybody and are commonly defined by all organizations: a **Malicious Software Control Policy** (an anti-virus policy), an **Acceptable Internet Usage Policy**, an **Acceptable Email Usage Policy**, a **Logical Access Control Policy**, a **Disaster Recovery (Backup) Policy**, a **Remote Access Control Policy**, and a **Third Party Access Control Policy**.

One requirement overrides all others at this level and is marked as critically important: the ISPA must **clearly indicate where each of these different policies originates**. In concrete terms, every sub-policy must be *linked* or *traced back* to the CISP in some way, so that the chain of authority from the board directive down to the individual policy is never broken. A well-formed sub-policy therefore carries an explicit **Reference** section pointing *up* to the statements in the CISP that give it origin, and *down* to the lower-level procedure that supports it, along with a **Main Policy Statement** and its supporting sub-statements. For example, an Internet Acceptable Use sub-policy would state that use of the company's Internet service must be in support of official company work, that all user requests for connectivity must be approved and supported by the employee's line manager and made through a specified application procedure, and that it is the line manager's responsibility to ensure the user does not misuse the service.

Procedures

At the bottom of the tree lies the **operational level**, inhabited by the **procedures**, which are effectively **algorithmic steps**. Where a sub-policy declares what must be true, a procedure specifies the exact ordered actions that make it true. In the running example, a user requesting Internet connectivity within the company triggers an application procedure that states precisely what must be done: the applicant records identifying details, signs a declaration that they have read both the CISP and the relevant Internet Acceptable Use sub-policy and will act in line with them, and obtains the signatures of the line manager and the information security officer. A procedure of this kind must also **clearly state what it means to be compliant with it**, carrying its own compliance clause so that conformance can later be checked.

Once the full set of documents is in place — stating, from top to bottom, what the organization needs to do — the procedures are what get implemented in the **Execution** part of the model. Execution is simply the carrying-out of those procedures, and the role of the Control process that follows is to verify that everything is in fact working as the documents require.

The Control Part and Compliance

Compliance as the Scope of Control

Once the documents of the Direct process exist and the procedures have been executed, the model turns upward into the **Control** part, whose main scope is the

compliance check. ISO 27002 makes this requirement explicit: after everything has been put in place, the organization must verify compliance at the various levels — compliance with external laws and contractual requirements, and compliance with the security policies and standards it has established internally. The standard devotes an entire clause to this, titled **Compliance**, with the objective of avoiding breaches of legal, statutory, regulatory, or contractual obligations related to information security and of any security requirements.

Two of its controls capture the essence of the task. The control on the **identification of applicable legislation and contractual requirements** demands that all relevant legislative, statutory, regulatory, and contractual requirements — and the organization's approach to meeting them — be explicitly identified, documented, and kept up to date for each information system and for the organization as a whole, with the specific controls and individual responsibilities also defined. The control on **compliance with security policies and standards** demands that managers regularly review the compliance of information processing and procedures within their area of responsibility against the appropriate security policies, standards, and other security requirements. ISO 27002 further requires that management initiate an **independent review**, carried out by individuals independent of the area under review — the internal audit function, an independent manager, or an external specialist — to ensure the continuing suitability, adequacy, and effectiveness of the organization's approach to managing information security; the results are recorded and reported back to the management that initiated the review, and if the approach is found inadequate, corrective actions must be considered.

The Measurability Problem

The central difficulty of the Control part is defining the **compliance criteria** — that is, deciding how to verify that the organization is in fact behaving correctly. This difficulty flows directly from the nature of the documents themselves. Policies, directives, and standards are written in **natural language**, and natural language is neither strict nor formalized; consequently it is genuinely hard to establish exactly what must be verified. The problem reduces to the governing maxim of the whole control process: **you can assess only what you can measure**. Things that cannot be measured cannot be controlled, so the organization must define measurements that make it possible to quantify how compliant it is. For this reason the compliance clauses, which encode these measurements, must provide a clear metric that can be evaluated and assessed.

Evaluating compliance is not equally hard at every level. At the **bottom** of the pyramid it is comparatively easy, because the operational layer deals in procedures and in concrete data drawn directly from the system, which can be monitored. If an access control policy declares that only authorized users may access a certain type of data, compliance can be checked simply by inspecting the system's logs. As one climbs **upward**, however, the criteria become far more general, and the natural question arises of what metric could possibly apply at those altitudes. The answer is the concept of **risk**. At the higher levels, verifying compliance means **assessing the risk that derives from non-compliance** with the policies and directives. The task, therefore, is to translate the technical data extracted from the environment into something that allows the risk level to be evaluated — to convert log entries and alert counts into a statement about exposure.

The Role of Time

In carrying out this exercise, **time plays a decisive and troublesome role**. Assessing risk is expensive in time for several compounding reasons. First, the necessary information is widely spread throughout the company and must be gathered from many places. Second, that data must then be analysed, which itself consumes time even when software is used to assist. Third, a conclusion must be derived from the analysis. Taken together, a full risk assessment can take **weeks or even months**. It cannot be performed daily; it is something that must be planned.

This temporal cost creates a structural tension with the domain itself. In cybersecurity, **weeks constitute a very large temporal frame** — the threat environment can change completely within such a window. Worse, the world is **dynamic**, so even if the organization manages to choose what seems like the right assessment interval, the context captured at the start of the assessment may no longer hold by the time the assessment concludes. Traditionally, compliance monitoring has been performed through periodic **ICT audit sessions**, in which ICT risks are evaluated, an audit report is produced, and its results are compared against the compliance clauses. The abiding issue with this approach is precisely timing: the world is dynamic, and monitoring ought ideally to be done in near real time, which the periodic-audit model struggles to deliver.

Guidelines for Writing Compliance Clauses

Unlike the Direct process, which comes equipped with templates and suggestions for writing the CISP and sub-policies, the Control process offers **no standardized**

content for compliance clauses — specifying them is not an exact science, and very few guidelines are available. Only a small number of principles can be stated. A compliance clause must be **clear and precise**, deliberately avoiding the insertion of ambiguity; otherwise two people performing the same assessment against the same clause might reach different conclusions simply because the criteria were open to interpretation. A clause such as "*Employees must take Information Security seriously*" is a textbook failure, because the word *seriously* has no fixed meaning and cannot be assessed. Beyond clarity, a compliance clause should **express a way to measure the degree to which it is satisfied**, providing a usable metric rather than a vague aspiration.

Worked Example — A Clause for the Board Directive

Consider an extract from a Board Directive in which the board accepts the importance of the company's IT infrastructure to its strategic goals, recognizes the potential impact a compromise could have, and consequently demands a proper environment to protect the IT infrastructure against any disruptions or compromises, fully supporting a proper ISG environment. Here the **asset** is the IT infrastructure and the **protection need** is protection against compromises. Two candidate compliance clauses illustrate the difference between a weak and a strong formulation.

Option 1 states that the board requires proper feedback, every three months, of the level of compliance with the directive. Its merit is the presence of a clear **time window**. Its defect is that the phrase *level of compliance* is never defined anywhere — it is left ambiguous exactly what is to be assessed. **Option 2** states that the board requires detailed feedback, every three months, of any serious risks that may compromise the protection of the company's ICT infrastructure. This option retains the same time window but is markedly better, because it also clarifies the **metric**: the thing to be reported is *risk*. Option 2 is therefore the superior clause.

Worked Example — Clauses for the CISP

The second example concerns the CISP, which, as established, is a document articulated as a collection of statements that act as pointers to sub-policies. Because of this structure, it is preferable to attach a compliance clause to **each statement** (or each category) rather than to the document as a whole. Taking Statement 6.1 — which declares that the company will have a proper information

security organizational structure to manage information security, ensuring that security roles are assigned, that users are aware of the policy and of the disciplinary consequences of non-compliance, and that implementation is continuously coordinated and reviewed — two candidate clauses can again be compared.

Option 1 states that the effectiveness of the company's information security organizational structure in managing and enforcing the requirements of the ISPA shall be evaluated every six months, with the results incorporated into higher-level reporting. **Option 2** states that the way the information security function is organized, and the reporting structures within that function, shall be evaluated every twelve months to ensure the structure remains adequate to protect all the company's electronic assets, with the results forming part of the annual evaluation and reporting of organizational structures. Both options share the strength of a defined **time frame**. Option 1's weakness is that it does not specify *what* must be reported, so a change in the person writing the report would produce a different view each time. Option 2 improves on this because it refers to a document that is already expected to be produced annually and will therefore most probably be structured in a consistent way — yet it, too, leaves the **metric** unexpressed, failing to say explicitly what is to be measured. The honest conclusion is that **no option here is perfect**, but Option 2 is better than Option 1. The recurring lesson across both examples is that a good clause pins down both a time window and a measurable metric, and that most real clauses succeed on one dimension while remaining imperfect on the other.

Risk Management

Risk as the Element That Drives Control

From the control point of view, the single element that makes the control loop implementable is **risk**. Risk, however, is perceived differently at every layer of the hierarchy, and to deal with it coherently the organization needs a dedicated process for its analysis: **Risk Management**. This process must be **vertical**, assessing risk at every level, and the closer it operates to the bottom of the pyramid, the more tightly it is bound to the technological layer.

Formally, risk management is the process of identifying and assessing all potential risks, as well as introducing the controls that should mitigate those risks to acceptably low levels. It is one of the security processes required to identify the risks that may impact the well-being of an organization. Its defining spirit is to

begin by looking for what might be dangerous, to establish to what extent it is dangerous, and then to seek possible solutions evaluated against a **cost–benefit principle** — the same cost-effectiveness discipline that governs the whole of ISG.

The Two Dimensions of Risk

Although there are several ways to perform the process, risk is generally understood through **two orthogonal dimensions**. The first is the **probability or frequency** — how likely it is that something bad will happen. The second is the **magnitude of gains or losses**, usually called the **impact** — how severe the consequence would be if it did happen. The aim of risk management is accordingly to determine two things: what the impact will be if the risk materializes, and how often the risk might materialize. These two dimensions combine into the canonical formula:

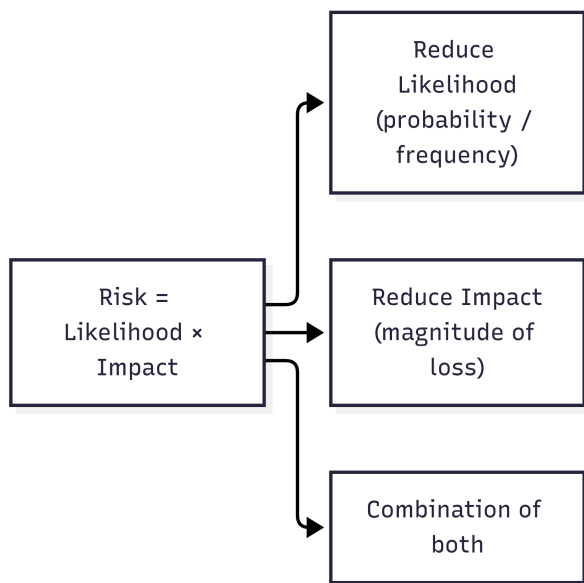
$$\text{Risk} = \text{Likelihood} \times \text{Impact} = (\text{Threat} \times \text{Probability}) \times \text{Impact}$$

The relationship is often displayed as a coloured matrix with likelihood on the vertical axis and impact on the horizontal axis, in which the combinations shade from green in the low–low corner, through amber in the middle band, to red where high likelihood meets high impact — a visual map that lets managers read off the severity of any risk at a glance from its coordinates.

The power of treating these dimensions as independent is best seen through examples. Suppose the incident of concern is the **loss of data**. One cause might be a **fire** destroying the data centre: with proper security protections in place the probability of this is quite low, yet should it occur it could destroy the centre completely, giving a very high impact. The result is a **critical risk even though the probability is low**. The same loss of data might instead arise from **ransomware**: here the probability is higher than that of fire, while the impact depends on the type of data affected and may well be smaller. This gives a risk that is again medium-to-high but with a smaller impact. Because the two scenarios occupy different regions of the likelihood–impact space, the appropriate **countermeasures differ** — the organization may choose to act one way against fire and an entirely different way against ransomware. Different models exist that decompose probability into more granular dominating factors, but in general the two governing quantities remain likelihood and impact.

How to Reduce Risk

The aim of the whole governance loop is to **reduce the level of risk**, and since risk is the product of likelihood and impact, there are exactly three levers available. One may **reduce the potential impact** of the risk; one may **reduce the probability or frequency** of the risk; or one may apply a **combination of both**. Which lever is pulled, and how hard, is itself a cost–benefit decision.



Risk Management, ISO 27002, and Governance

ISO 27002 treats IT risk analysis as an **essential part of best practice in IT**. It holds that an organization must identify its security requirements, and it names three main sources of such requirements. The first is the **assessment of risks** to the organization, taking into account its overall business strategy and objectives; through a risk assessment, threats to assets are identified, vulnerability and likelihood of occurrence are evaluated, and potential impact is estimated. The second is the body of **legal, statutory, regulatory, and contractual requirements** that the organization, its trading partners, contractors, and service providers must satisfy, together with their socio-cultural environment. The third is the organization's own set of **principles, objectives, and business requirements** for handling, processing, storing, communicating, and archiving information in support of its operations. The standard stresses that the resources spent implementing controls must be balanced against the business harm that would likely result in their absence — once again the cost-effectiveness balance — and it points to ISO/IEC 27005 for detailed guidance on risk assessment, treatment, acceptance, communication, monitoring, and review.

A structural principle closes the topic. All risks that could possibly have a negative effect on the well-being of the organization, once they materialize, are definitively the **responsibility of management**. It follows that **all levels of**

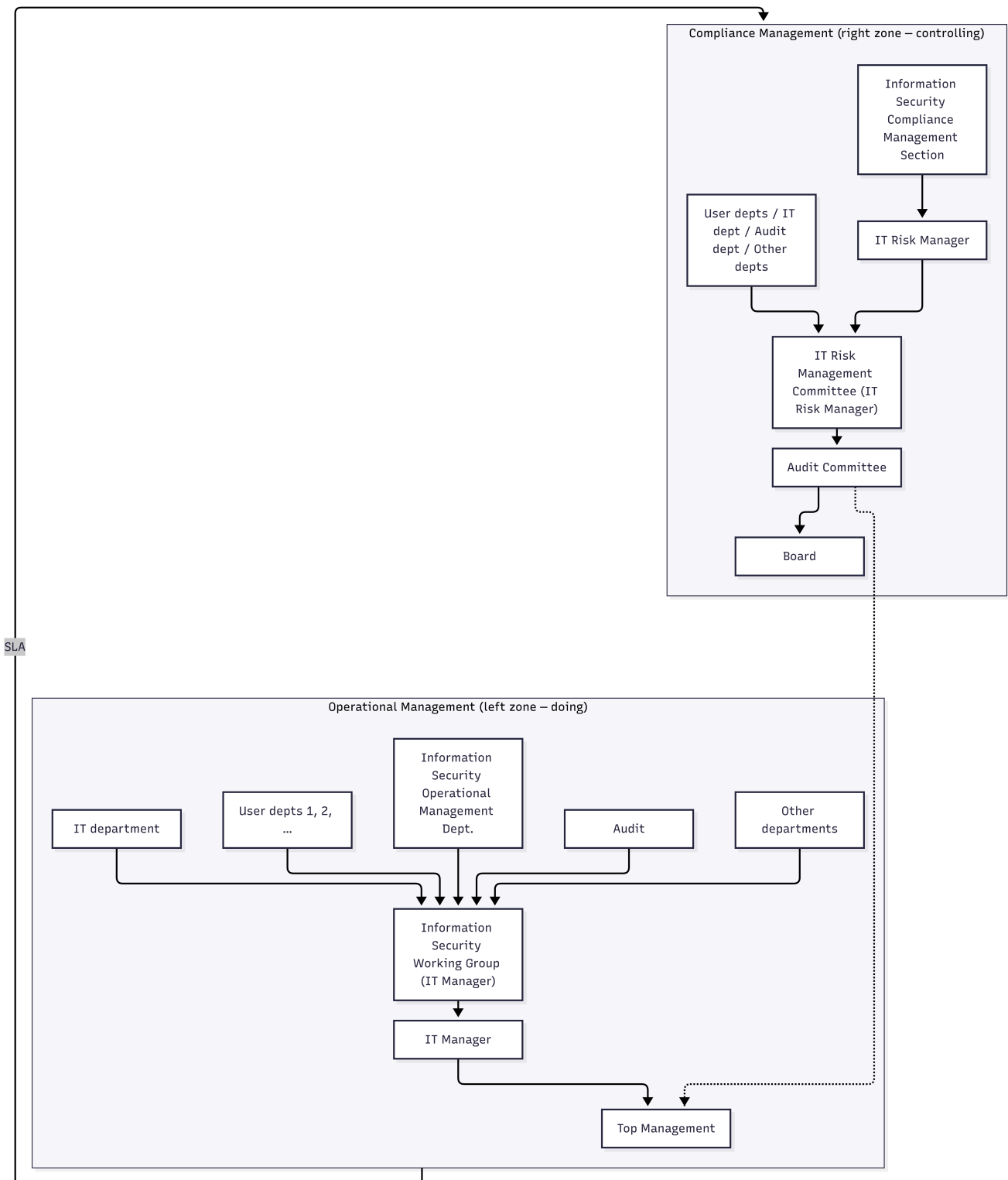
organizational management must be involved in the process of risk management, and that it is imperative for information and IT-related risks to be managed in an **integrated manner together with business risks** rather than in a technical silo. This is especially important in cybersecurity, because the people at the strategic level typically have no technical background; it therefore falls to the middle level to assess everything worth investigating, to highlight the critical issues, and to bring them to the attention of the managers who take decisions, since those managers must be made aware of all the potential bad situations that could occur.

Organization of Information Security

Two Orthogonal Security Functions

The way information security is organized inside a company is itself a matter of great importance, and every best practice document underlines this — ISO 27002, for instance, dedicates a clause titled *Organization of Information Security* to it. The guiding recommendation is that the organization should contain **at least two distinct components**, and that these two components are **orthogonal** to one another. The first looks after **day-to-day operational aspects** — operational security. The second is responsible for the **compliance monitoring function** — compliance management. The reason for the separation is structural: the people who *do* things must not be the same people who *check how things are done*, otherwise control would be grading its own work.

This produces a two-zone organizational chart. The **left zone**, Operational Management, is where the operational activities are actually carried out, implementing the bottom of the pyramid. The **right zone**, Compliance Management, is where the control process is implemented and feedback is provided back to the Direct process. The two zones meet at the top, at Top Management, and are connected to each other through a **Service Level Agreement (SLA)** that governs the relationship between what is operated and what is audited.



In the operational chart, an **Information Security Working Group** led by the IT Manager aggregates input from the user departments, the dedicated Information Security Operational Management department, the audit function, the IT department, and other departments, and reports upward through the IT Manager to Top Management. In the compliance chart, an **Information Security Compliance Management Section** feeds the **IT Risk Manager**, who sits on an **IT Risk Management Committee**; that committee collects input from all the departments, assembles a consolidated view — the matrix of how things are done — and passes it to an **Audit Committee** that reports to the Board. In

narrative terms, IT management on the operational side gathers information from the operational layer and composes it into a matrix view, which it hands to the person performing the audit; the audit side then collects input from across the departments and relays it to the risk management team, which in turn reports to top management.

Information Security Operational Management

The operational dimension is the set of people charged with ensuring that the system is functioning, organized into groups that monitor different departments, aggregate their information, and report to IT management. The activities it encompasses are the concrete, everyday actions of running security. They include **logical access control management**, meaning the real actions of adding to, changing, and deleting user access rights from access control lists. They include **identification and authentication management**, meaning the real actions of adding to, changing, and deleting entries in the user ID database and password files. They include **firewall management** — configuring firewalls with the authorized access rights, connecting workstations to LANs and to the Internet. They include **virus and malicious software management**, such as installing and updating antivirus software, together with the **handling of antivirus and related security incidents**. They include the **setting and updating of the security settings and configurations** of workstations and servers, the **ensuring of availability through UPS systems**, and the **ensuring of backups and the secure storage of those backups**, among other such day-to-day duties.

Information Security Compliance Management

The compliance dimension manages a different and orthogonal set of concerns, all directed at verifying that things are done in the right way and at assessing whether risk exists or persists. The activities that must be managed here include the level to which previously identified IT risks are managed and mitigated; the level of information security awareness of users; the availability, completeness, and comprehensiveness of the information security policies, procedures, and standards; the level of compliance to those policies, procedures, and standards; the impact on the company's IT risk level when policies are not complied with; the compliance with regulatory, legal, and statutory requirements; and software licensing issues, among others. In practice this side also takes the compliance clauses defined in the documents and attempts to apply them, verifying not only that

risks are controlled but that the whole documentary edifice is current, complete, and actually followed.

Awareness and the SETA Programme

Why Awareness Matters at Every Level

A recurring principle of the whole model is that **all levels are involved in the ISG process**, and awareness is the mechanism by which that involvement reaches the human core of the company. Information security procedures, guidelines, and practices must be drafted and conveyed to **all** users of information and IT in the organization, which means that every information worker must be made aware of — and trained in — the information security policy together with its associated procedures, guidelines, and practices. This is precisely why the board directive and CISP documents must be distributed as widely as possible: awareness is the capability of the company to carry a genuine security attitude, to recognize security issues, and to know how to handle them, and it cannot exist unless the knowledge is spread.

The SETA Programme and Its Three Temporal Horizons

To build this capability the model recommends developing a **Security Education, Training and Awareness (SETA)** programme — an instrument operating at the human level of the company, whose purpose is to inject the security factor into people. SETA should be an *extension* of the general knowledge employees already possess to do their jobs; its objective is to teach each of them to do that same job **securely**. The ambition is to move the awareness level from the basics to something that becomes automatic in the person's everyday life, creating a common understanding of both the issues and the solutions.

The programme has three stated objectives: to improve awareness of the importance and the need to protect organizational information resources; to acquire the necessary skills and know-how to do jobs more securely; and to create an understanding and insight into *why* it is important to protect organizational information assets. These three objectives map onto three distinct activities, each operating over a different time horizon, as captured by the differentiation between awareness, training, and education.

Awareness answers the attribute *what*, operates at the level of *information*, has the objective of making people *alert to danger*, is delivered through media such as videos, newsletters, and posters, and has a **short-term** impact — this is the first stage, developing the capability to understand what is dangerous. **Training** answers the attribute *how*, operates at the level of *knowledge*, has the objective of building *skill*, is delivered through practical instruction such as lectures, workshops, and hands-on practice, and has a **medium-term** impact — this is the stage that explains how to behave. **Education** answers the attribute *why*, operates at the level of *insight*, has the objective of *understanding*, is delivered through theoretical instruction such as seminars and literature study, and has a **long-term** impact — this is the most challenging stage, making people understand *why* they should behave in a certain way, and it is effective only over the long run.

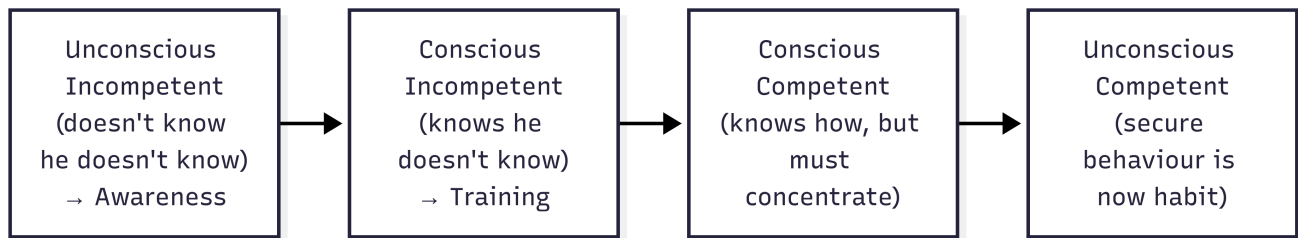
Attribute	Awareness	Training	Education
Attribute (question)	What	How	Why
Level	Information	Knowledge	Insight
Objective	Alert to	Skill	Understanding
Teaching method	Media (videos, newsletters, posters)	Practical instruction (lecture, workshop, hands-on)	Theoretical instruction (seminar, literature study)
Impact timeframe	Short-term	Medium-term	Long-term

The Conscious Competence Learning Model

The journey an employee takes through a SETA programme is described by the **Conscious Competence Learning Model**, a four-stage progression from ignorance to instinct.

The starting point is **Unconscious Incompetence**: the person does not even realize that they do not know how to do a certain thing safely. The priority here is to make the individual aware that they are incompetent as far as that specific task is concerned, and this is the province of **awareness** exercises. The second stage is **Conscious Incompetence**: the employee now knows that they lack the skill, and the priority becomes **training** them to do the job in a secure manner. The third stage is **Conscious Competence**: the employee knows what to do and how to do

it to keep the job secure, but still has to *concentrate* in order to perform the necessary procedures correctly. The fourth and final stage is **Unconscious Competence**: through enough practice and experience, all the security-related practices and skills have become inherent to the person's normal daily actions and behaviour — security has become a **habit** requiring no conscious effort.



The Pillars of Training

Underpinning the programme are the substantive topics that training must actually cover. At the level of *motivation*, employees must be taught why information is such an important asset, what the information security policy and procedures are, what the role and responsibility of every employee is, and what the consequences of not complying with the policy will be. At the level of *concrete practice*, the training should address user identification and authentication through passwords, how to choose a password, and the prohibition on sharing user IDs and passwords; virus control; backing up information; the sharing and storing of confidential information; secure email usage; secure Internet usage; social engineering; the handling of mobile devices such as notebooks and PDAs; the legal usage of software; office manners and discipline such as a clean-desk policy; and the required actions whenever a security incident or breach is suspected.

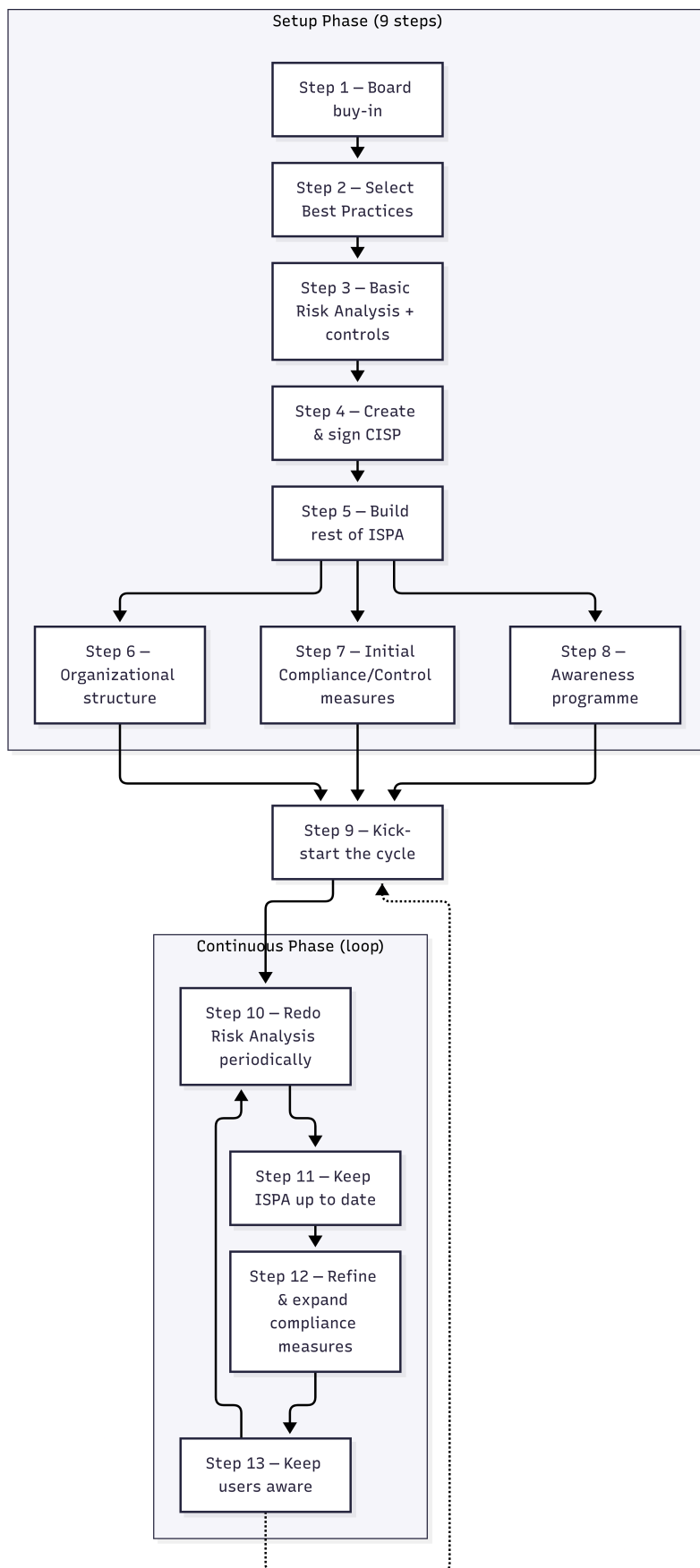
A Methodology for Establishing an ISG Environment

Putting the Whole Model Together

Having examined the Front Dimension, all five layers of the Depth Dimension, and the Best Practices basement, the final question is how to assemble the pieces into a working environment — *how do we put it all together?* The answer is a high-level **methodology of 14 steps**, deliberately written for the hardest case: it is a *complete* methodology that assumes no ISG environment exists at all and that everything must be started from scratch. In most real companies the situation is considerably better, so the methodology also carries a standing instruction to **re-use as much as possible** to avoid wasting time and money: the actual situation

must be compared against the steps, and must be synchronized and integrated with them wherever possible.

The 14 steps divide into two phases. The **Setup Phase** comprises the first nine steps (Steps 1 through 9), executed largely in sequence although Steps 6, 7, and 8 branch out in parallel from Step 5. The **Continuous Phase** comprises the final five steps (Steps 9 through 13, with Step 9 acting as the bridge that kick-starts the loop), which then repeat perpetually, implementing the Direct-Control flow on an ongoing basis.



The Setup Phase (Steps 1–9)

Step 1 — Get the Board's buy-in about IT risk management and information protection. This is a critically important starting point, and without it the project will certainly struggle to progress properly. It is acceptable that one cannot

initially obtain a *clean* Board Directive; what is essential is a genuine commitment from the board to support the project, which is enough to begin. Indeed, a basic, poor risk assessment may be performed first to identify the assets, under the approval of management, because the Direct process cannot start until it is known what must be protected. The formal Board Directive can come later — but it must be secured.

Step 2 — Select some guiding Best Practices. It is fundamental to choose an existing international best practice to serve as the foundation of the project and to provide motivation and direction. Many exist, and the essential point is simply to *have one*.

Step 3 — Perform a basic Risk Analysis and determine all needed controls. Some form of risk analysis is mandatory in order to identify the genuinely critical assets and to avoid wasting money and investment. This analysis requires involving a wide spectrum of people — board members, line management, users, system owners — and it must ensure that the relevant controls are installed and operational.

Step 4 — Create a CISP and have it signed by the Chairman/CEO. A CISP must be drafted, circulated among the stakeholders, and then submitted to top management. Recall that a more formal Board Directive on information security is essential in order to write a consistent CISP. The draft, once signed, becomes official company policy, and it forms the basis and motivation for all future steps.

Step 5 — Create the rest of the ISPA. The documents composing the full architecture must now be defined, taking as input the risk analysis from Step 3 and the CISP from Step 4, and following the guidelines discussed earlier. Throughout, the **connections between the documents must be preserved**, so that every sub-policy and procedure traces back up the tree.

Step 6 — Create the organizational structure for ISG. Specific attention must be paid to both the Operational Management and the Compliance Management sides, identifying the teams to be involved and the people responsible for them, and assigning responsibilities. Because this step is closely related to Step 7, it is preferable to perform the two **in parallel**.

Step 7 — Create an initial set of Compliance/Control measures and begin using them to produce reports on all three management levels. It is important here to obtain buy-in from the company's internal and external IT

audit departments as well as the legal department; if these were already involved in Step 3, the process is much easier. Creating such measures is not straightforward, and their effectiveness and value will have to be refined over time — the point is to start with an initial set that forms the basis of the Control part of the model.

Step 8 — Create an Awareness Programme, including aspects such as information security job responsibilities. This step is core to the success of the whole effort and must be performed on a continuous basis, with all the documents of the ISPA forming part of the programme.

Step 9 — Get the cycle going and kick-start the process. At this point the whole programme must be initiated; if one already exists, the revised programme must be integrated with the existing one to make the new one operational. Step 9 is the hinge between the two phases.

The Continuous Phase (Steps 10–13)

Step 10 — Redo the Risk Analysis from time to time to identify changes in risks and controls. Risks are dynamic: old ones disappear as patches are installed, ports are closed, and access controls are deployed; new ones materialize as new software is installed or configuration parameters change; and existing ones shift in impact, since a single mitigation action may decrease the impact of several other threats at once. Periodic reassessment keeps the risk picture current and the controls operational.

Step 11 — Keep the ISPA up to date and aligned with the newly identified risks, ensuring that all changed risks are reflected by updating the content of the ISPA and its compliance clauses where necessary.

Step 12 — Refine and expand the Compliance/Control measures to cater for newly identified risks, enforce compliance, and keep reporting to top management. The choice and creation of compliance clauses and monitoring measures remain challenging, and the process never stops: experience is gathered continuously and used to refine existing measures and to create new ones.

Step 13 — Continue to make all users more information-security aware. This process can never stop and must be enforced on a continuous basis, after which the loop returns to Step 10.